

MAE403 Computational Fluid Dynamics, Fall 2026

Homework Set 2: Understanding and running example codes; Taylor expansion

Due Monday, March 16, 2026 by 19:00, by online submission through *Blackboard*

5. (20 points) Download the code `Channel_FD_onwall0_class.f90` from the course Blackboard website and read the code.
 - (a) Using the lecture notes as a guide, explain what flow problem this code is trying to solve;
 - (b) Provide a sketch of the flow domain and the locations of the grid points;
 - (c) Describe, in your own words, all the input parameters for this code and their meaning;
 - (d) Describe, in your own words, what are the output data of the code?
6. (40 points) We shall solve the transient laminar flow problem numerically, by the following algorithm: the explicit Euler scheme in time, and central finite-difference in space,

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = g + \nu \frac{u_{j-1}^n - 2u_j^n + u_{j+1}^n}{\Delta y^2}, \quad (1)$$

where $\Delta y = 2L/N$, $u_j^n = u(y_j, t_n)$ with $y_j = -L + (j-1)\Delta y$ and $t_n = n\Delta t$ ($n = 0, 1, 2, 3, \dots$). The two end nodes $j = 1$ and $j = N+1$ are located on the channel walls. Assume $L = 1$, $u_0 \equiv gL^2/(2\nu) = 1$, and $\nu = 0.1$. The time step size will be set according to $\Delta t = 0.32\Delta y^2/\nu$.

- (a) Start with the code Dr. Wang provided in the lecture. Run the code with $N = 8, 16$, and 32 . Plot and compare the velocity profiles from different grid resolutions at $\nu t/L^2 = 1.28m/N^2 = 0.2, 1.0, 5.0$ (here m is the number of time steps. You need to modify Dr. Wang's code as needed. You also need to find a nice plotting package (*e.g.*, Matlab) to do the plots. Also find out at what dimensionless time, $\nu t/L^2$, the velocity at the center of the channel ($y = 0$) reaches the value of $0.97u_0$ (meaning the steady state is essentially reached) for the three different grid resolutions, respectively.
- (b) Use the analytical solution as the benchmark, plot the normalized error as a function of y/L , at $\nu t/L^2 = 0.2, 1.0, 5.0$, for the three grid resolutions in (a). Make your own observations. The normalized error could either be defined as

$$\frac{|u_{\text{numerical}} - u_{\text{analytical}}|}{u_{\text{analytical}}}, \quad (2)$$

or

$$\frac{|u_{\text{numerical}} - u_{\text{analytical}}|}{u_0}, \quad (3)$$

where $u_{\text{numerical}}$ and $u_{\text{analytical}}$ represent the numerical and analytical solution, respectively.

- (c) Plot and compare the time evolution of the properly normalized wall viscous stress from the three resolutions and the analytical solution. Explain how to compute the wall viscous stress for both the numerical and analytical solutions.
- (d) Summarize any problems you have encountered when doing this problem, and what you have learned after doing this problem (in numerical method, coding, plotting, accessing the department Unix server, Unix system, *etc.*).

7. (20 points) Show, by Taylor expansion, that

$$\frac{d^2 f(x_i)}{dx^2} = \frac{f_{i+1} - 2f_i + f_{i-1}}{\Delta x^2} - \frac{\Delta x^2}{12} \frac{d^4 f(x_i)}{dx^4} + \mathcal{O}(\Delta x^4) \quad (4)$$

For $f(x) = \sin(\pi x/2) + 0.1x^5$, what is the error in computing $d^2 f/dx^2$ at $x_i = 1$ with $\Delta x = 0.1$? How is the error compared to the leading-order estimate in Eq. (3)?

8. (10 points) For a time-dependent differential equation,

$$\frac{du(t)}{dt} = F(u(t), t), \quad (5)$$

the third-order Adams-Bashforth schemes is given as

$$\frac{u^{(n+1)} - u^{(n)}}{dt} = \alpha F(u^{(n)}, t^{(n)}) + \beta F(u^{(n-1)}, t^{(n-1)}) + \gamma F(u^{(n-2)}, t^{(n-2)}), \quad (6)$$

where dt is the time step size.

- (a) Using Taylor expansion, determine α , β , and γ , such that the scheme has the least truncation error.
- (b) What is the order of accuracy of the resulting scheme?